Can Vision Models Truly Read Charts?
A Multimodal Study of Chart Question Answering

Milan Addepalli
Department of Electrical and Computer Engineering, Rice University
ma200@rice.edu

Abstract

Charts convey information through an inherently different visual vocabulary than natural photos (i.e. through heights of bars, slopes of lines, proportions within a pie, etc.). However, most vision encoders are almost exclusively pretrained on natural photographic data, which instead contain real-life objects, such as people, animals, and places. We study whether these encoders can meaningfully understand charts through the task of Chart Question Answering (ChartQA). That is, given a chart image and a natural language question, predict the correct answer. We evaluate three systems on the ChartQA benchmark: a frozen CLIP model plus an MLP classifier, a CLIP model with an LoRA-adapter visual encoder plus an MLP classifier, and zero-shot evaluation with the Qwen2.5-VL-7B. Our results show an exceptionally clear percentage point gap between CLIP-based models agnostic of visual adaptation and SOTA VLMs like Qwen2.5. We apply GradCAM and Attention Rollout to both trained models, findings that both methods separately confirm CLIP attends diffusely across chart regions rather than localizing to task relevant features. Together, our quantitative and qualitative analyses reveal that the visual domain gap between natural photos and charts can't be bridged by lightweight adaptation of photo-pretained encoders.

1. Introduction

Charts and graphs are some of the most abundant and vital mediums for conveying quantitative information, appearing extensively in academic research, financial analysis, and journalism. The ability to answer natural language questions about these types of figures could be very useful in a variety of fields, yet it remains a surprisingly difficult task for modern vision-language systems.

The difficulty can be traced to a fundamental mismatch, which is that state-of-the-art visual encoders like CLIP [10] are pretrained on hundreds of millions of natural image-text pairs, learning visual representations that are optimized for photographic content. Charts, in contrast, communicate information through an entirely different visual vocabulary. The *height* of a bar, the *slope* of a line, and the *area* of a pie slice bear almost no resemblance to the visual cues that make photo-pretrained models effective. It is therefore far from obvious whether these model can "read" a chart, or whether visual domain adaptation is necessary.

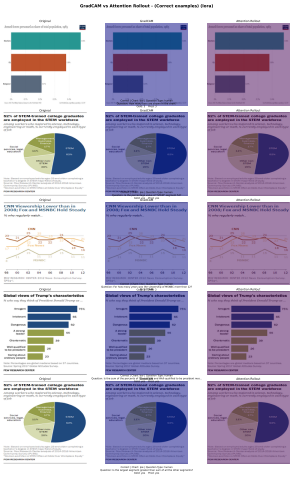

Figure 1. GradCAM and Attention Rollout visualizations for correct LoRA CLIP predictions across variety of charts. **Left:** original chart. **Center:** GradCAM. **Right:** Attention Rollout. Both methods separately produce broad, diffuse heatmaps with no spatial localization to task-relevant chart features, confirming perception bottleneck [7] despite the predictions being correct.

This project investigates that question utilizing the ChartQA benchmark [8]. We evaluate three systems: (1) a **frozen CLIP + MLP** classifier that tests how much chart understanding is already latent in pretrained representations, (2) a **LoRA CLIP + MLP** classifier that applies lightweight visual domain adaptation, and (3) **zero-shot Qwen2.5-VL-7B** [2] as an upper-bound baseline. Beyond accuracy, we perform explainability analysis using GradCAM [11] and Attention Rollout [1] to understand *why* models succeed or fail.

2. Related Works

Chart Understanding. ChartQA [8] provides real-world chart images with human-authored and machine-generated question answer pairs. Prior synthetic benchmarks include FigureQA [5], DVQA [4], and PlotQA [9]. Importantly, Liu et al. [7] show that CLIP achieves near random image-to-text retrieval accuracy on ChartQA, which directly motivated our investigation of CLIP as a chart understanding backbone.

Vision Language Models. CLIP [10] is trained through contrastive learning on 400 million image-text pairs. While it transfers well across a lot of visual domains, its representations for specialized visual languages such as charts are limited [7]. Recent general purpose VLMs, such as LLaVA [6] and Qwen2.5-VL [2], are trained on broad multimodal instruction data including documents and figures, enabling strong zero-shot performance on charts.

Parameter-Efficient Fine-Tuning. LoRA [3] does a reparameterization of the updates that are applied to the training weights as low-rank matrix products, which allows for more efficient fine-tuning. This method was originally proposed for large language models, but has also since been applied to stable diffusion models and vision transformers. We apply LoRA to CLIP's visual encoder attention layers to see whether targeted adaptation can help improve chart understanding in a noteworthy enough way.

Interpretability. GradCAM [11] produces class-discriminative saliency maps from gradients that flow into a target layer. Attention Rollout [1] recursively computes the token attentions across all transformer layers to gain the collective attention from the CLS (classification) token at the beginning of the input sequence to each patch. Utilizing both methods provides intersecting evidence when they agree.

3. Method

3.1. Architecture

Given a chart image I and question q, the visual encoder f_V and text encoder f_T produce normalized embeddings $\mathbf{v}, \mathbf{t} \in \mathbb{R}^d$. These are then concatenated and passed through a two-layer MLP to obtain the logits over the answer vocabulary:

$$\hat{y} = W_2 \, \mathrm{ReLU}(W_1, [\mathbf{v}; \mathbf{t}] + b_1) + b_2 \tag{1}$$

We use CLIP ViT-B/32 ($d = 512$), MLP hidden dimension on 512, and dropout $p = 0.3$. The task is a closed-vocabulary classification over the top-$K = 5{,}001$ most frequent training answers, which cover 73.7% of gold answers.

3.2. Training Variants

Frozen CLIP: both encoders are frozen in this case and only the MLP is trained ($\sim$3.1 million parameters). **LoRA CLIP:** low-rank adapters are applied to the query and value projection matrices of the visual encoder's attention layers. Each target projection W is replaced by:

$$W' = W + \frac{\alpha}{r} BA, \quad A \in \mathbb{R}^{r \times d_{\text{in}}}, \; B \in \mathbb{R}^{d_{\text{out}} \times r} \tag{2}$$

with rank $r = 8$ and scaling $\alpha = 16$ using the heuristic $\alpha = 2r$. Only A, B, and the MLP head are updated, adding 294 thousand trainable parameters, which is only 0.34% of the visual encoder.

3.3. Zero-Shot Baseline

Qwen2.5-VL-7B-Instruct is queried with a short prompt asking for a concise answer, using greedy decoding. No task-specific training is performed. This tests what a large general-purpose VLM achieves through just instruction following.

3.4. Interpretability Analysis

GradCAM is applied to the last visual encoder block, weighting patch activations by gradient of the predicted class score with respect to each feature activation and applying ReLU. Attention Rollout hooks every attention layer, then recursively multiples attention matrices with residual connections ($A' = 0.5A + 0.5I$) to obtain collective attention starting from the CLS token and through all patches. Both maps are upsampled to image resolution through bilinear interpolation and normalized to $[0, 1]$.

4. Experiments

Dataset. ChartQA [8] contains 28,299 training, 1,920 validation, and 2,500 test samples. Questions are split into *human-authored* and *augmented* types. From the official annotation files, 84% of the charts are bars, 12% are lines, and 4% are pie types.

Evaluation Metric. We report *relaxed accuracy* [8]. With this specific metric, if both the prediction and gold (groundtruth) values are both numeric, then the prediction is correct if it's within 5% of the gold value. This is to account for data extraction inaccuracies. Otherwise, it's just like regular accuracy where a prediction is correct if it's an exact match to the gold answer. We stratify results by question types, chart type, and answer type (binary, numerical, or textual).

Training Details. Both trained models use AdamW as the optimizer with a learning rate of 3×10^{-4} and weight decay of 10^{-4}. They are trained for 20 epochs with a batch size

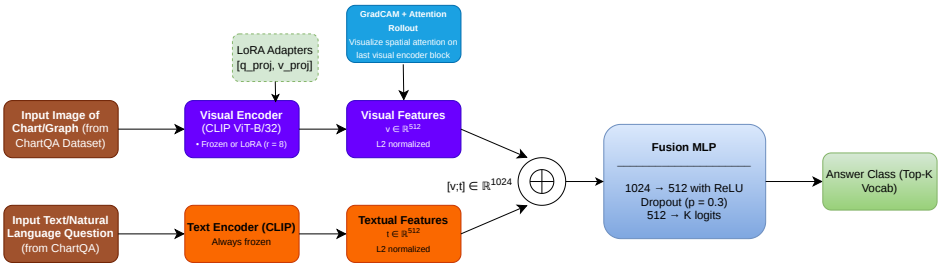

Figure 2. Model architecture. Visual and text encoders are pretrained CLIP components. For the LoRA variant, low-rank adapters are injected into the query and value projection matrices of the visual encoder. The MLP fusion head is always trained from scratch.

of 64 and cosine annealing to gradually decrease learning rate during training following a cosine function. The best checkpoint is selected by validation accuracy. Both models plateau by epoch 5, indicating the bottleneck in the visual features rather than training duration.

5. Results and Discussion

Table 1. Relaxed accuracy (%) on ChartQA test set by model variant, question type (H=human, A=augmented), and chart type (B=bar, L=line, P=pie.)

Model	Overall	H	A	B	L	P
Frozen CLIP	4.9	7.4	2.4	4.1	8.5	5.2
LoRA CLIP	5.4	7.1	3.6	4.4	9.5	7.0
Qwen2.5-VL	**68.3**	**62.3**	**74.2**	**70.4**	**67.3**	**46.5**

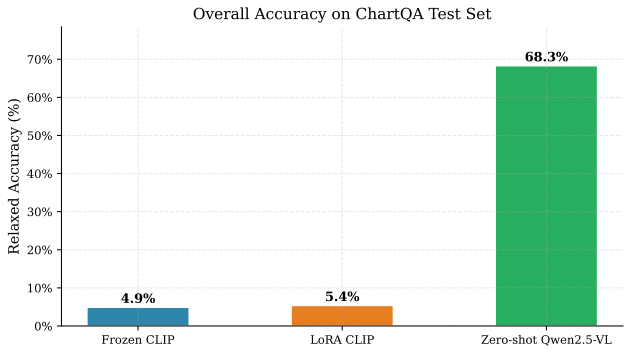

Figure 3. Overall relaxed accuracy. The central finding is the 63 percentage point gap between both CLIP models and Qwen2.5-VL.

The gap in chart understanding is profound. The CLIP-based models tested achieve only 4.9-5.4% overall accuracy versus Qwen2.5-VL's 68.3% (Table 1, Figure 3). This is consistent with Liu et al. [7], who show that CLIP achieves near-random retrieval on ChartQA, suggesting its visual embeddings fail to encode chart-discriminative information in a linearly accessible way (e.g. dot product, cosine similarity, linear classifier).

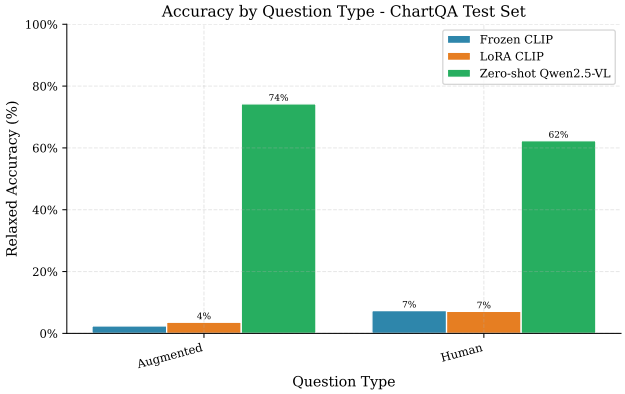

Figure 4. Accuracy by question type. CLIP models perform better on human questions while the Qwen2.5-VL follows the reverse pattern.

LoRA barely helps. LoRA adaptation improves accuracy by only $+0.5$ percentage point (4.9% *to* 5.4%). Adapting 0.34% of the visual encoder's parameters is insufficient to bridge the chart understanding gap. This is consistent with Liu et al. [7], who find that meaningful improvement requires chart-specific contrastive retraining at scale rather than lightweight fine-tuning.

Human vs. augmented reversal. CLIP models perform better on human questions (7.1–7.4%) than augmented ones (2.4–3.6%), while Qwen2.5-VL shows the opposite (62.3% human, 74.2% augmented). Augmented questions require precise numerical answers, which is extremely challenging for a closed-vocabulary classifier but pretty straightforward for a generative model. Human questions involve more common textual answers that appear in the top-5000 vocabulary, explaining CLIP's relative advantage there.

Answer type stratification Stratifying by gold answer type reveals that nearly all correct CLIP predictions fall on binary (yes/no) or common textual answers, with near-

zero accuracy on numerical questions. The model defaults to high-frequency vocabulary items under uncertainty — "yes" and "no" are the top-2 vocabulary entries (339 and 255 training occurrences) — producing the systematic pattern of binary predictions for numerical questions visible in qualitative examples.

Chart type patterns. Line charts are consistently easier than bar charts for CLIP models (human×line: 11% vs. human×bar: 7%), likely because line charts have fewer data points and simpler spatial structure. For Qwen2.5-VL, pie charts are dramatically harder (35–48%) than bar or line charts (62–76%), suggesting that spatial proportion reasoning is a speciifc weakness even for large VLMs.

GradCAM and Attention Rollout confirm the perception bottleneck. Figure 1 shows three-column visualizations for correct LoRA CLIP predictions. Both GradCAM and Attention Rollout separately produce broad, diffuse heatmaps with no spatial localization to task-relevant features despite the predictions being correct. Their agreeement is particularly significant because the two methods have fundamentally different theoretical foundations. GradCAM measures gradient flow (what the model *used* to predict) while Attention Rollout measures structural information flow (where the model *looked*). When both show the same diffuse pattern, it strongly suggets CLIP truly lakcs spatially grounded chart understanding rather than merely failing to surface it through one analysis method. Frozen and LoRA heatmaps are nearly identical, consistently with the minimal accuracy gap between the two variants.

6. Conclusion

We investigated whether photo-pretrained vision models can understand charts through the ChartQA task. CLIP-based classifiers achieve only 4.9–5.4% relaxed accuracy regardless of LoRA adaptation, while zero-shot Qwen2.5-VL achieves 68.3%. Both GradCAM and Attention Rollout independently confirm that CLIP attends diffusely across chart regions, providing visual evidence of the perception bottlebekc identified by Liu et al. [7]. Our findings suggest that bridging the visual domain gap between natural photographs and charts requires either large-scale multimodal pretraining on document and figure data, or strenous chart-specific contrastive adaptation of the visual encoder. Lightweight fine-tuning by itself is insufficient.

References

[1] S. Abnar and W. Zuidema. Quantifying attention flow in transformers. *arXiv preprint arXiv:2005.00928*, 2020.

[2] S. Bai, K. Chen, X. Liu, et al. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025.

[3] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. Lora: Low-rank adaptation of large language models. *ICLR*, 2022.

[4] K. Kafle, B. Price, S. Cohen, and C. Kanan. Dvqa: Understanding data visualizations via question answering. In *CVPR*, 2018.

[5] S. E. Kahou, V. Michalski, A. Atkinson, et al. Figureqa: An annotated figure dataset for visual reasoning. In *ICLR Workshop*, 2018.

[6] H. Liu, C. Li, Q. Wu, and Y. J. Lee. Visual instruction tuning. In *NeurIPS*, 2023.

[7] J. Liu, W. Zeng, X. Zhang, Y. Wang, Z. Shan, and J. He. On the perception bottleneck of vlms for chart understanding. *arXiv preprint arXiv:2503.18435*, 2025.

[8] A. Masry, D. X. Long, J. Q. Tan, S. Joty, and E. Hoque. Chartqa: A benchmark for question answering about charts with visual and logical reasoning. In *Findings of ACL*, 2022.

[9] N. Methani, P. Ganguly, M. M. Khapra, and P. Kumar. Plotqa: Reasoning over scientific plots. In *WACV*, 2020.

[10] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al. Learning transferable visual models from natural language supervision. In *ICML*, 2021.

[11] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *ICCV*, 2017.